Product Requirements Document on TradeSentinel

Team013 (Individual) · Jodhdeep Singh · 2025mb26112 · MBA – AI in Business · Management of AI Products · November 2026

Status: Draft for pilot review. **Owner:** Jodhdeep Singh. **Scope:** order-entry and execution issues; one client tier; one venue.

1. Goal and Non-Goals

- **Goal:** reach a correct first triage faster, with near-zero critical under-triage, and with evidence attached.
- **Not in this release:** other flows; L3 automation; any production action; any client-facing bot; model training on client data.

2. Users

User	Context	Needs	Error cost they carry
L1/L2 support engineer	Handles the queue during market hours, under time pressure	Clear reading of the report; quick routing; no blame for hidden AI errors	A missed fault; a wasted hour on the wrong path
Subject matter expert (SME)	Checks unclear, failed-trade, security and critical issues	Only real escalations; full evidence on arrival	Alert fatigue if over-escalation is high
Development team	Applies approved fixes through the change process	A clear cause, evidence and suggested fix	Rushed or wrong fixes in production
Operations and risk lead	Owns SLA and automation decisions	Evidence for gates; audit trail; quick rollback	Reportable events; regulatory exposure

3. User Stories, Including Uncertainty

ID	Story	Acceptance criteria
US1	As an engineer, I see the AI's reading, its confidence and alternatives.	Category, confidence and next-best options are shown on every ticket above L0.
US2	When the system is unsure, I want to see that and have an SME check it.	Below 55% confidence: "unsure" marker, strictest plausible floor, SME check.
US3	I want to see why severity is what it is.	A severity trail lists each step and reason.
US4	I want duplicate reports grouped without losing control.	Suggested group shown; merging needs a person; blast radius can raise severity.
US5	I want to disagree and have that count.	Every change requires a reason, which is logged.
US6	I want likely causes with proof.	Each cause lists at least one past incident or release.
US7	I do not want to send a client something unproven.	The draft can be edited; Send is locked while any blank remains; no cause appears without a citation.
US8	As a risk lead, I want to change automation without a release.	Levels are set per category in configuration; rollback is immediate.
US9	As an engineer, I want to see at a glance which issues need me and which are waiting on the client.	Home board has two lanes; sending an update moves an issue to the client lane with what is awaited and since when; a client reply moves it back.

4. Functional Requirements

ID	Requirement	Priority
FR1	Classify report into one of six categories with calibrated confidence	Must
FR2	Propose severity; apply category floors by market hours; never lower it	Must
FR3	Apply security rule and blast-radius rule	Must
FR4	Route by automation level: L0 hidden; L1 suggestion; L2 pre-filled	Must
FR5	Suggest duplicates by similarity and time window	Must
FR6	Log model version, suggestion, severity steps, decision, reason and review time	Must
FR7	Rank causes with evidence; show runbook; assign the suggested fix to the development team	Should
FR8	Draft an editable client reply and a closure note, with citations and visible blanks	Should
FR9	Show issues in two lanes by who must act next, with counts; keep detail behind toggles	Must

ID	Requirement	Priority
FR10	L3 auto-routing for low-risk categories	Later, after Gate 3

5. Never Allowed at Any Level

Downgrade severity; close an issue; merge issues; deploy or roll back in production; send an unapproved update; state an uncited cause.

6. Control Configuration

Category	Owner	Floor: market / off hours	Level in MVP
Order rejected	Order Gateway Support	S3 / S4	L2
Order stuck (no confirmation)	Order Management (OMS)	S2 / S3	L2
Execution failed	Execution Engineering	S2 / S3, always SME check	L1 (held on purpose)
Trade or position mismatch	Post-Trade Engineering	S3 / S4	L2
Connection problem	Connectivity Support	S2 / S3	L2
Access or how-to question	Client Service Desk	S4 / S4	L2

Other rules: security terms set at least S2 and route to SME + Security. Three linked reports set at least S2; four in market hours set S1. Levels in the MVP show the mechanism only; they have not been earned, and the pilot starts all categories at L0.

7. Data Assumptions

- Incident records hold usable cause and fix text.
- Release logs carry timestamps and component names.
- Identifiers can be masked before any API call without losing meaning.
- Historic severity labels are unreliable, so experts relabel a sample.

8. Fallback

- **Model unavailable or slow:** ticket drops to the manual queue, marked “no AI reading”; rules that do not need the model continue to run.
- **No similar incident:** the assistant says so and does not guess; the runbook remains available.
- **Metric breach:** the category drops a level by configuration.

9. Open Questions

- Which API route satisfies data and contract requirements?
- What is the real share of critical issues?
- Who owns control configuration after launch?
- Which events are reportable, and does the audit log meet that need?